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Comprehensive study on image forgery techniques using deep learning

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***Abstract*— Image forgery detection using deep learning is a technique that utilizes artificial neural networks to identify whether an image has been morphed or manipulated. This is an important task in the digital age, as the proliferation of easily accessible photo editing software has made it easy for individuals to alter images for various purposes, such as spreading misinformation or altering evidence. Deep learning algorithms work by training on large datasets of images and learning to recognize patterns and features that are indicative of forgery. These algorithms can then be used to analyze new images and determine whether they have been manipulated. One approach to image forgery detection using deep learning is to train a convolutional neural network (CNN) on a dataset of both genuine and forged images. The CNN can then be used to classify new images as either genuine or forged based on the features it has learned to recognize. An additional strategy is to make use of a generative adversarial network, often known as a GAN. This system is made up of two neural networks that collaborate with one another to generate and categorise images. By training, the GAN on a dataset of genuine and forged images, the discriminator network can learn to recognize the features of forged images and classify them accurately. There are several challenges to overcome when using deep learning for image forgery detection. Despite these challenges, deep learning has shown promising results for image forgery detection and is an active area of research. In the future, it is likely that deep learning algorithms will play an important role in detecting and preventing image forgery, helping to maintain the integrity and trustworthiness of digital images.**

Keywords*— *Image forgery, Deep learning, CNN, ANN.

I. Introduction

Deep learning is a technique utilized to detect image forgery. This process, which is commonly used in digital image editing, can be done through the use of artificial neural networks. The rise of software that can easily be used to alter photos has made it easier for people to do so. Due to the accuracy of deep learning algorithms when it comes to detecting forgery, the use of this technology has gained widespread attention. A deep learning algorithm can be used to identify patterns and features in images that are characteristic of forgery. It can then analyze new images to determine if they have been digitally altered. There are various approaches to detecting forgery using deep learning[1], [2].

One method to identify forged or genuine images is by using a CNN, which is a type of neural network. To train the CNN, it is required to analyze a dataset of both forged and genuine images. It then learns the various features that are associated with forgery and classifies new images according to these features. According to studies, CNNs are very effective at detecting image tampering, such as changes in the spatial domain. Another method is to use a generator and a discriminator network, which are two neural networks that are designed to create and classify images[3]. The former creates synthetic images while the latter attempts to distinguish them from real ones. Through training the GAN on an image dataset, the discriminator can learn to recognize the various features of forged and genuine images. A GAN can also be used to detect image forgery, particularly when it comes to identifying changes in the frequency domain[4], [5].

A third approach is to use a hybrid approach that combines the strengths of both CNNs and GANs. For example, a CNN can be used to identify tampering in the spatial domain, while a GAN can be used to identify tampering in the frequency domain. By combining the outputs of both networks, it is possible to achieve improved accuracy in detecting forged images. There are several challenges to overcome when using deep learning for image forgery detection. One challenge is the need for a large dataset of both genuine and forged images to train the algorithms. It can be difficult to obtain a sufficient number of forged images, as they are typically created by individuals with the intention of misleading or deceiving others. Another challenge is that deep learning algorithms can be fooled by sophisticated forgeries that are difficult for humans to detect. This is known as adversarial attack, where the forger intentionally creates a forged image that is specifically designed to fool the deep learning algorithm[5].

Despite these challenges, deep learning has shown promising results for image forgery detection and is an active area of research. In the future, it is likely that deep learning algorithms will play an important role in detecting and preventing image forgery, helping to maintain the integrity and trustworthiness of digital images. One potential application of image forgery detection using deep learning is in the field of journalism, where accurate and reliable images are critical for maintaining the credibility of news articles. Deep learning algorithms could be used to automatically scan images in news articles and flag those that have been tampered with, alerting editors to the potential for misinformation. Another potential application is in the field of criminal investigations, where images may be used as evidence in court. Deep learning algorithms could be used to automatically analyze images and identify those that have been tampered with, ensuring that only reliable and trustworthy images are presented as evidence[6], [7].

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II. RELATED WORK

Hongwei Yao et al.[8] presents a method for detecting and localizing image forgeries using a reliability fusion map. The proposed method first extracts features from an input image using a convolutional neural network (CNN). These features are then fed into multiple classifiers, each of which makes a prediction about the authenticity of the image. The classifiers are trained to make predictions based on different types of image forgery, such as copy-move, splicing, and resampling. The predictions of the classifiers are combined using a fusion map, which is a graphical representation of the confidence of each classifier in its prediction. The fusion map is used to identify the regions of the image that are most likely to be forged, allowing for localization of the forgery. The authors of the paper evaluate the performance of the proposed method on a dataset of forged images and show that it outperforms several state-of-the-art image forgery detection methods. The proposed method is also able to localize the forgeries with high accuracy.

Neeraj Kumar et al.[9] presented a singular value decomposition to detect image forgeries (SVD). The method divides the input image into overlapping blocks and applies SVD to each block. SVD can decompose a matrix into its components, revealing its key features. The authors use SVD to identify image block forgeries by analysing singular values and singular vectors from the decomposition. They propose comparing the singular values and singular vectors of image blocks to detect copy-move forgeries..

Ankit Kumar Jaiswal et al.[10] proposes a multi-scale, multi-stage deep learning model to detect digital image copy-move forgery. Two stages—feature extraction and classification—comprise the model. The model extracts robust and discriminative features from the input image using LBPs and SIFT features. To capture image content variations, SIFT features are extracted at multiple scales. A CNN classifies extracted features in the classification stage. The CNN learns complex copy-move forgery patterns and features from a large dataset of forged and genuine images.

Fatima Zahra El Biach et al.[11] a method that uses an image forgery detection system composed of a CNN and an encoded-decoder algorithm. The first step is to extract the various features of the input image, while preserving its most important ones. The CNN then trained to extract the lower-dimensional elements while preserving the main features of the image. After the extracted features have been fed into a CNN decoder-based program, it can then reconstruct the input image. This method can help determine if the input image has been manipulated or altered. The proposed method was tested on a set of genuine and forged images. The results show that it can detect image forgery at high accuracy. It can also detect various types of forgery, such as copy-move and resampling.

Ritu Agrawal et al.[12] proposed a method involves extracting features from the input image using a deep learning-based feature extraction algorithm. The extracted features are then used to identify regions of the image that have been copied and pasted within the same image, by searching for matches between the features in different regions of the image. To improve the efficiency of the method, the author proposes using a sliding window approach to search for matches, which allows for the detection of forgery in large images without the need for exhaustive searches. The proposed method is tested on a dataset of forged and genuine images, and the results show that it is able to achieve high accuracy in detecting copy-move forgery. The method is also able to detect forgery in a variety of different types of image manipulation, including splicing, resampling, and compression.

Ali Taimori et al.[13] proposes detecting digital image double JPEG compression. Double JPEG compression—saving an image in JPEG format twice—can degrade image quality and introduce artefacts. The proposed method analyses the image's co-occurrence matrix and pixel intensity distribution. These properties can detect double JPEG compression and estimate degradation. The proposed method is tested on a dataset of images with different levels of double JPEG compression, and the results show that it can accurately detect and estimate the level of degradation.

Syed Sadaf Ali et al.[4] proposes deep learning and image recompression to detect image forgery. First, JPEG is used to recompress the input image. A deep learning-based classifier uses the recompressed image's features to distinguishbetween real and fake images. Image recompression and deeplearning can detect copy-move, splicing, and resampling forgeries, according to the authors. The proposed method detects image forgery accurately on a dataset of forged and genuine images.

U. Kosarkar et al.[15] proposes face recognition with two well-known image recognition methods such as Principal Component Analysis (PCA) and K- Nearest Neighbour(KNN). The proposed methods improve the face recognition to extract features from the face image and also reduce the dimensionality of each image. Also used K-Nearest Neighbour(KNN) to classify accurate data.

Image forgery detection method

There are several methods that can be used to detect image forgery, which refers to the act of altering or manipulating digital images in an attempt to deceive the viewer. Some of the common methods for detecting image forgery include:

- Analysis of image metadata: Image metadata includes information about the camera, settings, and other details related to the image capture process. By analyzing the metadata, it is possible to determine if the image has been manipulated or altered.
- Statistical analysis: Statistical analysis can be used to detect anomalies or inconsistencies in the image data that may indicate that the image has been manipulated. For example, analyzing the distribution of pixel intensities in the image can reveal patterns that are characteristic of image manipulation.
- Feature extraction and matching: Feature extraction algorithms can be used to extract unique characteristics from the image, such as edges, corners, or textures. These features can then be used to search for matches between different regions of the image, which can reveal if the image has been manipulated by copying and pasting regions of the image.

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- Deep learning: Deep learning algorithms, such as convolutional neural networks (CNNs), can be trained to recognize patterns and features that are characteristic of image manipulation. By feeding an input image into a trained CNN, it is possible to detect if the image has been manipulated.

Some major image forgery detection methods are discussed below:

Digital watermarking

Digital watermarking is a technique used to embed a digital signal or pattern into a digital image, audio, or video file in order to identify ownership or authenticity. This technique can be used to detect image forgery by comparing the watermarked image with the original image to determine if any changes have been made. There are several types of watermarking techniques, including visible watermarking, where the watermark is visible on the image, and invisible watermarking, where the watermark is not visible but can be detected through specialized software or methods. To implement a digital watermarking system, the original image is first watermarked using a specific algorithm and then stored in a secure location as shown in fig.1. When the watermarked image is later retrieved, it is compared to the original image to determine if any changes have been made. If the watermarked image has been altered in any way, it will not match the original image and can be detected as a forgery. Digital watermarking is a useful tool for detecting image forgery, but it is not foolproof. Watermarks can be removed or altered, and sophisticated forgers may be able to create fake images that pass the watermarking test. It is important to use multiple methods for detecting image forgery in order to ensure the authenticity of an image.

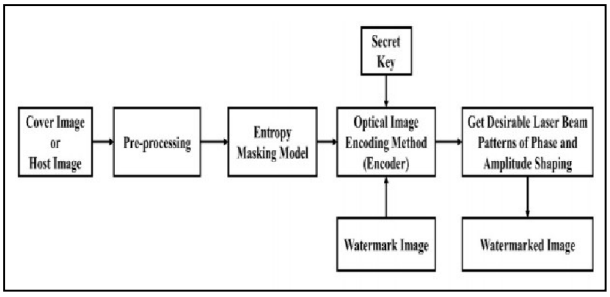

Fig. 1 Digital watermarking

Digital Signature

The identity of the sender of a digital message or document can be verified with the help of a digital signature, which is a specific kind of electronic signature. They can be used to detect image forgery by comparing the digital signature on the image with the original digital signature to determine if any changes have been made. To implement a digital signature system for image forgery detection, the original image is first signed using a specific algorithm and then stored in a secure location. When the signed image is later retrieved, it is compared to the original image to determine if any changes have been made. If the signed image has been altered in any way, it will not match the original image and can be detected as a forgery. Digital signatures are a useful tool for detecting image forgery, but they are not foolproof. Digital signatures can be forged or altered, and sophisticated forgers may be able to create fake images that pass the digital signature test. It is important to use multiple methods for detecting image forgery in order to ensure the authenticity of an image.

Image retouching

Image retouching is the process of altering or enhancing the appearance of a digital image using image editing software. It is a common technique used in photography and graphic design to improve the aesthetics of an image. However, it can also be used to create fake images or to alter the appearance of an existing image in a way that is not immediately detectable. To detect image forgery through image retouching, it is important to carefully examine the image for any signs of manipulation. This can include looking for inconsistencies or anomalies in the image, such as unrealistic lighting or shadows, unnatural coloration, or areas of the image that do not match the surrounding pixels. There are also specialized tools and techniques that can be used to detect image retouching, such as error level analysis (ELA), which compares the error levels of different areas of the image to determine if any changes have been made. Other methods include analyzing the metadata of the image, which can reveal information about the editing history of the image, and using image analysis algorithms to detect changes or inconsistencies in the image.

Fig. 2 Image retouching a) original image b) tampered image [14]

Splicing method

Splicing is a technique used to create a fake image by combining or manipulating multiple images or elements from different sources. It can be difficult to detect spliced images because they often appear to be seamless and realistic. To detect image forgery through splicing, it is important to carefully examine the image for any signs of manipulation or inconsistencies. This can include looking for irregularities in the image, such as unrealistic lighting or shadows, unnatural coloration, or areas of the image that do not match the surrounding pixels. There are also specialized tools and techniques that can be used to detect spliced images, such as analyzing the metadata of the image, which can reveal information about the editing history of the image, and using image analysis algorithms to detect changes or inconsistencies in the image. Overall, detecting image forgery through splicing requires a combination of careful examination and the use of specialized tools and techniques. It is important to use multiple methods for detecting image forgery in order to ensure the authenticity of an image.

Copy move method

The copy-move method is a technique used to create a fake image by copying and pasting elements from one part of an image to another. It can be difficult to detect copy-move forgeries because they often appear to be seamless and

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realistic. To detect image forgery through the copy-move method, it is important to carefully examine the image for any signs of manipulation or inconsistencies. This can include looking for irregularities in the image, such as unrealistic lighting or shadows, unnatural coloration, or areas of the image that do not match the surrounding pixels. There are also specialized tools and techniques that can be used to detect copy-move forgeries, such as analysing the metadata of the image, which can reveal information about the editing history of the image, and using image analysis algorithms to detect changes or inconsistencies in the image. One common technique for detecting copy-move forgeries is to divide the image into overlapping blocks and compare the blocks to each other to see if any of them are identical. If identical blocks are found, it may indicate that the image has been tampered with using the copy-move method.

Forensic Approach

Forensic techniques can be used to detect image forgery in a variety of ways, including:

1. Examining the metadata of the image: Image metadata contains information about the image, such as the date and time it was taken, the camera and lens used, and any editing history. Analyzing the metadata can help to reveal any inconsistencies or anomalies that may indicate the image has been tampered with.
2. Analyzing the pixel data: The pixel data of an image contains information about the color and intensity of each pixel in the image. Analyzing the pixel data can help to detect any irregularities or inconsistencies that may indicate the image has been tampered with.
3. Examining the physical characteristics of the image: The physical characteristics of an image, such as the resolution, color depth, and file size, can be used to identify any inconsistencies or abnormalities that may indicate the image has been tampered with.
4. Using specialized tools and techniques: There are a variety of specialized tools and techniques that can be used to detect image forgery, such as error level analysis (ELA), which compares the error levels of different areas of the image to determine if any changes have been made, and image analysis algorithms that can detect changes or inconsistencies in the image.

Deep learning approach in image forgery

Deep learning is a type of machine learning that involves the use of artificial neural networks to learn and make decisions. It has been applied to a variety of tasks, including image forgery detection. There are several approaches to using deep learning for image forgery detection, including:

- Training a deep learning model on a large dataset of both original and forged images and using the model to classify new images as either original or forged.

- Using a generative model to create a synthetic version of the original image and comparing it to the input image to determine if any changes have been made.

- Training a deep learning model to generate a "fingerprint" or signature for each image, and comparing the fingerprints of the input image to the original image to determine if any changes have been made.

Overall, deep learning can be a powerful tool for detecting image forgery, but it is important to carefully design and train the models and to use multiple methods for detecting image forgery in order to ensure the authenticity of an image

Various deep learning algorithm for detecting image forgery

There are many different deep learning algorithms that can be used for image forgery detection. Some of the most common ones include:

- Convolutional Neural Networks (CNNs): These are commonly used for image classification tasks and have also been applied to image forgery detection. CNNs can learn features from the input images and use them to classify whether an image is genuine or forged.
- Recurrent Neural Networks (RNNs): These are useful for tasks that involve sequential data, such as natural language processing and time series analysis. RNNs have been used for image forgery detection by analyzing the temporal consistency of an image over time.
- Generative Adversarial Networks (GANs): GANs consist of two networks: a generator and a discriminator. The generator produces fake images, while the discriminator tries to classify whether an image is real or fake. GANs have been used for image forgery detection by training thediscriminator to identify forged images.

It is important to note that no single algorithm is likely to be the best for all types of image forgery. The choice of algorithm will depend on the specific characteristics of the forged images and the goals of the detection system.

Implementation of deep learning

Here in a fig.3 that illustrates the general process for detecting image forgery using deep learning:

- **Preprocessing**: The input image is preprocessed to prepare it for analysis. This may include resizing the image, converting it to a specific color space, or applying other image processing techniques.

- **Feature extraction**: The preprocessed image is then passed through a feature extraction network, which extracts relevant features from the image. This may include techniques such as convolutional neural networks (CNNs), which can identify patterns and features within the image.

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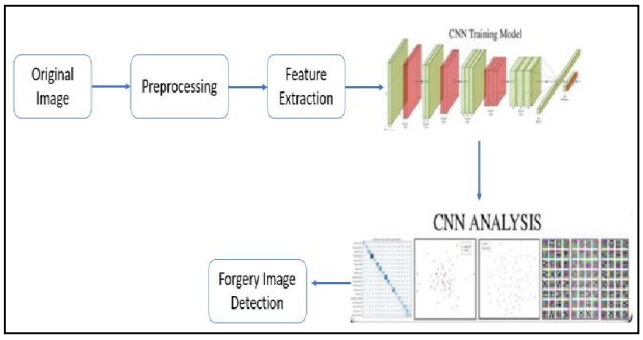

Fig. 3 CNN Analysis

- Classification: The extracted features are then passed to a classification network, which uses them to classify the image as either genuine or forged. The classification network may be trained on a dataset of genuine and forged images, allowing it to learn the characteristics that distinguish one from the other.

- Output: The classification network produces an output indicating whether the input image is genuine or forged. This output can be used to alert users or take other appropriate actions.

III. CONCLUSION

Image forgery detection using deep learning is a promising approach to detecting the presence of tampering or manipulation in digital images. There are various techniques that can be used for this purpose, including the use of convolutional neural networks (CNNs) to analyze the pixel values and patterns in an image, as well as techniques that leverage the inherent properties of digital images, such as the noise patterns and compression artifacts that are present in different types of images. Overall, the use of deep learning for image forgery detection is a rapidly evolving field, and there are many promising approaches and techniques that are being developed and refined. As the capabilities of machine learning algorithms continue to improve, it is likely that we will see even more sophisticated and effective methods for detecting image forgeries in the future

REFERENCES

[1] Z. J. Barad and M. M. Goswami, "Image Forgery Detection using Deep Learning: A Survey," *2020 6th Int. Conf. Adv. Comput. Commun. Syst. ICACCS 2020*, pp. 571–576, 2020, doi: 10.1109/ICACCS48705.2020.9074408.

[2] G. Kaur, N. Singh, and M. Kumar, *Image forgery techniques: a review*, no. May. Springer Netherlands, 2022.

[3] U. Kosarkar, G. Sakarkar, and S. Gedam, "An Analytical Perspective on Various Deep Learning Techniques forDeep Fake Detection," *Int. J. Innov. Eng. Sci.*, vol. 7, no. 8, pp. 25–30, 2022, doi: 10.46335/ijies.2022.7.8.5.

[4] P. Gupta, C. S. Rajpoot, T. S. Shanthi, D. Prasad, A. Kumar, and S. S. Kumar, "Image Forgery Detection using Deep Learning Model," *3rd Int. Conf. Smart Electron. Commun. ICOSEC 2022 - Proc.*, pp. 1256–1262, 2022, doi: 10.1109/ICOSEC54921.2022.9951952.

[5] D. Gragnaniello, F. Marra, G. Poggi, and L. Verdoliva, "Analysis of adversarial attacks against CNN-based image forgery detectors," *Eur. Signal Process. Conf.*, vol. 2018-Septe, pp. 967–971, 2018, doi: 10.23919/EUSIPCO.2018.8553560.

[6] H. Chen, C. Chang, Z. Shi, and Y. Lyu, "Hybrid features and semantic reinforcement network for image forgery detection," *Multimed. Syst.*, vol. 28, no. 2, pp. 363–374, 2022, doi: 10.1007/s00530-021-00801-w.

[7] B. Diallo, T. Urruty, P. Bourdon, and C. Fernandez-Maloigne, "Robust forgery detection for compressed images using CNN supervision," *Forensic Sci. Int. Reports*, vol. 2, no. September 2019, p. 100112, 2020, doi: 10.1016/j.fsir.2020.100112.

[8] H. Yao, M. Xu, T. Qiao, Y. Wu, and N. Zheng, "Image forgery detection and localization via a reliability fusion map," *Sensors (Switzerland)*, vol. 20, no. 22, pp. 1–18, 2020, doi: 10.3390/s20226668.

[9] N. K. Rathore, N. K. Jain, P. K. Shukla, U. S. Rawat, and R. Dubey, "Image Forgery Detection Using Singular Value Decomposition with Some Attacks," *Natl. Acad. Sci. Lett.*, vol. 44, no. 4, pp. 331–338, 2021, doi: 10.1007/s40009-020-00998-w.

[10] A. K. Jaiswal and R. Srivastava, "Detection of Copy-Move Forgery in Digital Image Using Multi-scale, Multi-stage Deep Learning Model," *Neural Process. Lett.*, vol. 54, no. 1, pp. 75–100, 2022, doi: 10.1007/s11063-021-10620-9.

[11] F. Z. El Biach, I. Iala, H. Laanaya, and K. Minaoui, "Encoder-decoder based convolutional neural networks for image forgery detection," *Multimed. Tools Appl.*, vol. 81, no. 16, pp. 22611–22628, 2022, doi: 10.1007/s11042-020-10158-3.

[12] R. Agarwal and O. P. Verma, "An efficient copy move forgery detection using deep learning feature extraction and matching algorithm," *Multimed. Tools Appl.*, vol. 79, no. 11–12, pp. 7355–7376, 2020, doi: 10.1007/s11042-019-08495-z.

[13] A. Taimori, F. Razzazi, A. Behrad, A. Ahmadi, and M. Babaie-Zadeh, *A novel forensic image analysis tool for discovering double JPEG compression clues*, vol. 76, no. 6. Multimedia Tools and Applications, 2017.

[14] T. Thakur, K. Singh, and A. Yadav, "Blind Approach for Digital Image Forgery Detection," *Int. J. Comput. Appl.*, vol. 179, no. 10, pp. 34–42, 2018, doi: 10.5120/ijca2018916108.

[15] U. Kosarkar, P. Sasankar, "A Study for Face Recognition Using Techniques PCA and KNN," IOSR Journal of Computer Engineering (IOSR-JCE) 3,2021.

[16] U. Kosarkar, Gopal Sakarkar, Shilpa "Revealing and Classification of Deepfakes Videos Images using a customize Convolutional Neural Network Model", https://doi.org/10.1016/j.procs.2023.01.237

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